

Xingyi Zhou

AMI Labs
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Google Scholar (18k+ citations)
Github (14k+ total stars)

Employment

- 2026– **AMI Labs**, *New York, NY*,
Member of Technical Staff.
- 2025–2026 **Meta Super Intelligent Lab**, *Bellevue, WA*,
Research Scientist.
- 2022–2025 **Google DeepMind**, *Kirkland, WA*,
Research Scientist.

Education

- 2017–2022 **The University of Texas at Austin**, *Austin, TX*,
Department of Computer Science, Ph.D.
Advisor: Dr. Philipp Krähenbühl
Thesis: Towards unified object recognition in the wild
- 2013–2017 **Fudan University**, *Shanghai, China*,
School of Computer Science, Bachelor in Science.

Research Experience

- 2018–2022 **Research Assistant**, *Deep Learning Lab, UT Austin*, Austin, TX,
with Dr. Philipp Krähenbühl.
Object detection; object tracking; general object representation.
- 2021 **Research Intern**, *Facebook AI Research*, Remote,
with Dr. Ishan Misra.
Large-vocabulary object detection.
- 2019–2021 **Research Intern**, *Intel Intelligent Systems Lab*, Santa Clara, CA/ Remote,
with Dr. Vladlen Koltun.
Multi-object tracking; object detection on multiple datasets.
- 2018 **Software Engineering Intern**, *Google Research*, Mountain View, CA,
with Tyler Zhu and Dr. Kevin Murphy.
Human pose estimation; human mesh reconstruction.
- 2017–2018 **Research Assistant**, *Graphics & AI Lab, UT Austin*, Austin, TX,
with Dr. Qixing Huang.
3D computer vision; pose estimation.
- 2014–2017 **Research Assistant**, *Institute for Media Computing, Fudan Univ.*, Shanghai, China,
with Dr. Wei Zhang.
Hand pose estimation; fine-grained image classification; decision trees.
- 2016 **Research Intern**, *Microsoft Research Asia (MSRA)*, Beijing, China,
with Dr. Yichen Wei.
Human/ hand pose estimation.

Selected Publications (Full list at [Google Scholar](#))

- 2025 **Gemini 2.5: Pushing the Frontier with Advanced Reasoning, Multimodality, Long Context, and Next Generation Agentic Capabilities**, arXiv, 2025, [[Paper](#)].
- 2025 **Visual Lexicon: Rich Image Features in Language Space**, XuDong Wang[†], Xingyi Zhou, Alireza Fathi, Trevor Darrell, Cordelia Schmid ([†]intern hosted at Google), CVPR, 2025, [[Paper](#)].
- 2025 **VoCap: Video Object Captioning and Segmentation from Any Prompt**, Jasper Uijlings*, Xingyi Zhou*, Xiuye Gu*, Arsha Nagrani, Anurag Arnab, Alireza Fathi, David Ross, Cordelia Schmid (* Equal contribution), arXiv, 2025, [[Paper](#)].
- 2025 **Dense Video Object Captioning from Disjoint Supervision**, Xingyi Zhou*, Anurag Arnab*, Chen Sun, Cordelia Schmid (* Equal contribution), ICLR (spotlight), 2025, [[Paper](#)] [[Code](#)].
- 2024 **Streaming Dense Video Captioning**, Xingyi Zhou*, Anurag Arnab*, Shyamal Buch, Shen Yan, Austin Myers, Xuehan Xiong, Arsha Nagrani, Cordelia Schmid (* Equal contribution), CVPR, 2024, [[Paper](#)] [[Code](#)].
- 2024 **Pixel Aligned Language Models**, Jiarui Xu[†], Xingyi Zhou, Shen Yan, Xiuye Gu, Anurag Arnab, Chen Sun, Xiaolong Wang, Cordelia Schmid ([†] intern hosted at Google), CVPR, 2024, [[Paper](#)] [[Code](#)].
- 2023 **How can objects help action recognition?**, Xingyi Zhou, Anurag Arnab, Chen Sun, Cordelia Schmid, CVPR, 2023, [[Paper](#)] [[Code](#)].
- 2022 **Detecting Twenty-thousand Classes using Image-level Supervision**, Xingyi Zhou, Rohit Girdhar, Armand Joulin, Philipp Krähenbühl, Ishan Misra, ECCV, 2022, [[Paper](#)] [[Code](#)].
- 2022 **Global Tracking Transformers**, Xingyi Zhou, Tianwei Yin, Vladlen Koltun, Philipp Krähenbühl, CVPR, 2022, [[Paper](#)] [[Code](#)].
- 2022 **Simple multi-dataset detection**, Xingyi Zhou, Vladlen Koltun, Philipp Krähenbühl, CVPR, 2022, [[Paper](#)] [[Code](#)].
- 2021 **Probabilistic two-stage detection**, Xingyi Zhou, Vladlen Koltun, Philipp Krähenbühl, arXiv, 2021, [[Paper](#)] [[Code](#)].
- 2021 **Multimodal Virtual Point 3D Detection**, Tianwei Yin, Xingyi Zhou, Philipp Krähenbühl, NeurIPS, 2021, [[Paper](#)] [[Code](#)].
- 2021 **Center-based 3D Object Detection and Tracking**, Tianwei Yin, Xingyi Zhou, Philipp Krähenbühl, CVPR, 2021, [[Paper](#)] [[Code](#)].
- 2020 **Tracking Objects as Points**, Xingyi Zhou, Vladlen Koltun, Philipp Krähenbühl, ECCV, 2020 (spotlight), [[Paper](#)] [[Code](#)].
- 2019 **Objects as Points**, Xingyi Zhou, Dequan Wang, Philipp Krähenbühl, arXiv, 2019, [[Paper](#)] [[Code](#)].
- 2019 **Bottom-up Object Detection by Grouping Extreme and Center Points**, Xingyi Zhou, Jiacheng Zhuo, Philipp Krähenbühl, CVPR, 2019, [[Paper](#)] [[Code](#)].
- 2018 **StarMap for Category-agnostic Viewpoint and Keypoint Estimation**, Xingyi Zhou, Arjun Karpur, Linjie Luo, Qixing Huang, ECCV, 2018, [[Paper](#)] [[Code](#)].
- 2018 **Unsupervised Domain Adaptation for 3D Keypoint Estimation via View Consistency**, Xingyi Zhou, Arjun Karpur, Chuang Gan, Linjie Luo, Qixing Huang, ECCV, 2018, [[Paper](#)] [[Code](#)].

- 2017 **Towards 3D Human Pose Estimation in the Wild: A weakly-supervised Approach**, Xingyi Zhou, Qixing Huang, Xiao Sun, Xiangyang Xue, Yichen Wei, ICCV, 2017, [[Paper](#)] [[Code](#)].
- 2016 **Deep Kinematic Pose Regression**, Xingyi Zhou, Xiao Sun, Wei Zhang, Shuang Liang, Yichen Wei, ECCV workshops, 2016, [[Paper](#)] [[Code](#)].
- 2016 **Model-based Deep Hand Pose Estimation**, Xingyi Zhou, Qingfu Wan, Wei Zhang, Xiangyang Xue, Yichen Wei, IJCAI, 2016, [[Paper](#)] [[Code](#)].

--- Horner

- Apr 2021 **Facebook Fellowship**.
- Jan 2020 **Facebook Fellowship Finalist**.
- Nov 2019 **Adobe Fellowship Finalist**.
- Sep 2017 **Provost's Graduate Excellence Fellowship**, UT Austin.
- Jun 2017 **Shanghai Outstanding Graduate**.
- Aug 2016 **Award of Excellence**, Stars of Tomorrow Internship Program, Microsoft Research Asia.

--- Competition Awards

- June 2021 **2nd place**, CVPR 2021 Waymo real-time 3D detection challenge.
- Aug 2020 **1st place**, ECCV 2020 Robust vision challenge, object detection track.
- Aug 2020 **1st place**, ECCV 2020 Robust vision challenge, instance segmentation track.
- Jun 2020 **2nd place/ Best student submission**, ICRA 2020 nuScenes 3D detection challenge.
- Dec 2014 **Gold medal**, ACM International Collegiate Programming Contest (**ACM-ICPC**) Regional.
- Nov 2012 **First prize**, National Olympiad in Informatics in Provinces (**NOIP**), Zhejiang Provinces.

--- Presentations

- Oct 2023 **Recognizing objects in long time and in a large-vocabulary**, ECCV 2022 MOTComplex workshop.
- Jan 2021 **Center-based 3D object detection and tracking**, VALSE Webinar [[Video](#)].
- Aug 2020 **Learning a unified label space for robust object detection**, ECCV 2020 Robust vision challenge workshop [[Video](#)].
- Aug 2020 **CenterNet2**, ECCV 2020 Joint COCO and LVIS Challenge workshop [[Slides](#)].
- May 2019 **Generalized keypoint estimation**, UT research preparation exam [[Slides](#)].
- Mar 2018 **Modeling geometry in pose estimation**, seminar presentation at UT Austin [[Slides](#)].
- Jan 2018 **Towards 3D human pose estimation in the wild**, GAMES Webinar [[Video](#)].

--- Professional Services

Conference Reviewer, ECCV(2018-), CVPR(2019-), ICCV(2019-), ICLR(2020-), NeurIPS(2020-). Outstanding reviewer in CVPR2019, ECCV2020, CVPR2021, CVPR2024.

Journal reviewer, TPAMI, TIP, IJCV, CVIU.

Teaching assistance, [Neural Networks](#), Fall 2019.

Web master and student host, UT [Forum for Artificial Intelligence](#).

Open-source softwares, [CenterNet](#) (7k+ stars), [CenterTrack](#) (2k+ stars), [Detic](#) (1k+ stars), [ExtremeNet](#) (1k+ stars), [CenterNet2](#) (1k+ stars), [pytorch-pose-hg-3d](#) (500+ stars).

Supervising experiences

- 2025 Kaiwen Zha, intern at Meta Super Intelligent Lab, PhD student from MIT
- 2024 XuDong Wang, intern at Google DeepMind, PhD student from UC Berkeley
- 2023 Jiarui Xu, intern at Google Research, PhD student from UC San Diego
- 2020 Tianwei Yin, undergraduate student from UT Austin
- 2018 Arjun Karpur, master student from UT Austin

Skills

- Languages Python, C/ C++, Lua
- Frameworks Jax (Codebases from scratch for Vision-Language Models, Segment Anything, and Object detection), PyTorch, Tensorflow, Torch, Caffe